Strategy versus Reality in a Leakage Free Preprocessing Pipeline for Malaria Risk Prediction in Rwanda

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Abstract

We implement and audit a leakage free preprocessing and evaluation pipeline that predicts a positive malaria rapid diagnostic test (RDT) from household, demographic and climate covariates in the 2023 Rwanda Malaria Indicator Survey (MIS). Every planned Phase 1 decision is compared with what the data forced in Phase 2 and backed by a number and a figure; eight decisions changed, including the missing code convention, the treatment of structural missingness and the winsorizing rule for zero spread features. The matrix grows from $18{,}672 \times 53$ to $14{,}940 \times 95$ (train) with no missing cells. Positives are rare (0.96%) and concentrated in a few clusters, so a 36 positive hold out is unreliable. Evaluating every one of the 179 positives out of fold under cluster grouped cross validation, a regularized logistic regression on household features plus cluster mean context reaches PR-AUC 0.050 (5.2 times the 0.0096 baseline) and ROC-AUC 0.72, whereas boosted trees on the 2020 climate covariates reach only 0.015 to 0.021. Interview month climate features built from the newly downloaded CHIRPS v3 rainfall, ERA5-Land dewpoint and MODIS EVI and NDVI did not improve the result (PR-AUC 0.011 to 0.021), and the report states why. Wide intervals and the data still needed are reported. The original project proposal is reproduced in Appendix A.

1 Introduction

Malaria remains a public health concern in Rwanda even after large reductions in transmission [1]. The task is binary: for a person who was tested, predict RDT positivity (`hml35`) so that bed nets and spraying can be targeted. Phase 1 fixed a preprocessing plan before any data were seen. Phase 2 executes it in Python inside `scikit-learn` `Pipeline` and `ColumnTransformer` objects [2] and records where reality disagreed. All statistics below are computed on training data only unless stated. Source code, figures and notebook are in the repository behind the GitHub Link under the title.

2 Data and Pipeline Output Verification

Data. The primary dataset is the Person Recode of MIS 2023 with cluster level geospatial covariates (5 km rasters, 2020 values) joined by cluster [3]. The file has 19,038 rows; 366 have no valid test and are removed (the target is never imputed), leaving $N = 18{,}672$ with $P = 53$ features (26 continuous, 22 nominal, 4 discrete, 1 ordinal). Overall missingness is 1.77% of cells; 9,933 rows are complete and 8,739 have at least one missing value.

Split. A cluster grouped, label stratified split keeps every survey cluster on one side only (cluster overlap $= 0$): train $14{,}940$ rows with 143 positives (0.957%, imbalance 103.5:1), test $3{,}732$ rows with 36 positives (0.96%). The split seed (51) was chosen by label balance only, never by model performance (seed 42 gave only 20 test positives).

Dimension verification. Table 1 reports the matrix before and after the pipeline. No rows are deleted by outlier handling, and no information is discarded: all 53 raw features are retained and expanded to 95 model columns (ordinal and one hot encoding of 22 nominal columns, plus missing indicator flags). The correlation and VIF screen is applied *only* on the linear model branch and removes 10 climate or terrain features, giving 85 columns there. PCA is used for diagnostics only; raw interpretable features are kept for classification.

3 Strategy versus Reality Decision Log

Table 2 lists each decision with the Phase 1 plan, the executed step and the evidence.

Stage	Rows	Cols	NaN cells
Raw model table	18,672	53	1.77%
Train, preprocessed	14,940	95	0
Test, preprocessed	3,732	95	0
Linear branch (after screen)	14,940	85	0

Table 1: Pipeline output verification (initial vs final $N \times P$). Train and test have identical column order; parameters are fit on train.

4 Empirical Justification

4.1 Missingness (Fig. 1, 2)

Only `hml4` and `hml10` exceed the 1% threshold (46.12% each); their missingness is exactly the households with no bed net. Little's test on the numeric block [4] rejects MCAR ($\chi^2 = 6984.3$, df $= 29$, $p < 0.001$, two patterns), and Cramér's V of the missing indicator with other variables is 1.00 (`hml12`), 0.70 (`hv227`), 0.36 (province), 0.17 (wealth), 0.06 (residence) and 0.04 (sex). The value is absent because of a skip rule, so imputing it by the mean or median would invent bed nets. Figure 2 shows the 103.5:1 imbalance, which fixes the metric choice (PR-AUC, not accuracy).

4.2 Outliers (Fig. 3)

`Nightlights_Composite` has skew 3.66 and kurtosis 13.89 (Q1 $= 0$, Q3 $= 0.095$, fences $[-0.142, 0.237]$) and 3,046 flagged values (20.39%). Age is mildly skewed (0.87; fences $[-33, 79]$; 124 values, 0.83%) and rooms per household has 718 flagged values (4.81%). Capping keeps N intact, which matters with 143 training positives. The `Frost_Days_2020` case is a genuine deviation: with Q1 $=$ Q3 $= 0$ the fence is $[0, 0]$ and the plain rule would remove all non-zero information, so the winsorizer uses percentile bounds when IQR is zero.

4.3 Association and collinearity (Figs. 4–6)

Twenty pairs reach $|r| \geq 0.85$, the strongest being mean temperature with PET ($r = 0.993$), maximum temperature (0.989) and minimum temperature (0.983), and precipitation with aridity (0.981). Before screening, 12 of 30 numeric features have VIF above 10 (maximum finite VIF about 4.5×10^{12} for maximum temperature, 1.8×10^5 for mean temperature). The screen drops 10 features (nine by the $|r|$ rule, maximum temperature by VIF $= 35.6$) and the maximum VIF falls to 7.54. Mutual information with the target is small for all features (top: land surface temperature 0.0162, wet days 0.0158, diurnal range 0.0156 nats). Because the rule keeps the feature with the larger $|r|$ to the target, land surface temperature (highest MI) was dropped; ranking by MI would have kept it, a documented limitation. Tree models are not screened. Multiple tests are corrected with Benjamini–Hochberg [5]; rankings are used for screening, not inference, because clustered data make naive p values optimistic.

4.4 Transformations and PCA (Figs. 7, 8)

For `Livestock_Goats`, log1p reduces skew from 4.12 to -0.19 (mean 90.1 to 4.25, sd 79.3 to 0.73); min-max gives mean 0.53, sd 0.15; standardization gives mean 0, sd 1. PCA on the 30 scaled numeric features has leading eigenvalues $11.58, 4.14, 2.52, 1.75, 1.56$ (variance $38.6\%, 13.8\%, 8.4\%, 5.8\%, 5.2\%$); 12 components reach 90% and 15 reach 95%. This shows strong redundancy among climate variables, but raw features are retained for interpretability.

5 Leakage Free Architecture

All steps are inside `Pipeline` and `ColumnTransformer` objects (sentinel recoder, winsorizers, imputers with indicators, log1p, scaler, rare level grouping, one hot and ordinal encoders, optional collinearity screen). Training skewness decides each column's branch, and every threshold, median, mean, standard deviation and capping bound is learned from `X_train` only and applied unchanged to test data. Cross validation uses cluster grouped, label stratified folds. Calibrators and operating thresholds are fit on out-of-fold predictions. The second malaria test, test-process variables, modelled malaria columns, anemia and treatment variables, identifiers are excluded as leakage, and the survey weight is used only in evaluation, never as a feature.

6 Model Results, Diagnosis and Improvement

Hold-out results. Candidate models are regularized logistic regression, random forest, LightGBM [6] and XGBoost [7], all with class weights; probabilities are calibrated with Platt scaling [8]. Table 3 gives the 20% cluster grouped hold-out. XGBoost has the best hold-out PR-AUC (0.0203; weighted 0.0193; out-of-fold 0.0455), but with 36

Step	Phase 1 plan	Phase 2 actual	Evidence and figure
Survey and unit	MIS 2017 or DHS 2019–20, children only	MIS 2023, all ages, age as a feature; MIS 2017 kept as secondary	Under 5 years gives only 24 positives of 2,412; MIS 2023 has GPS covariates, 2017 has none
Missing codes	Sentinel 9 recoded to NaN	Codes 3, 4, 6 for `hml35` (6, 7 for `hml32`) recoded; 366 rows without valid test dropped	Value counts of the real file; code 9 never occurs
Eligibility filter	Filter on `hv042`=1 and `hc1` age	`hv042` is hemoglobin selection and `hc1` is absent in PR, so filter on valid `hml35` instead	Codebook inspection
Missingness mechanism	Test MCAR/MAR, then impute	`hml4`, `hml10` are missing by design (46.12%, 6,891 of 14,940); explicit "no net" level plus indicator, no imputation	Little's MCAR $\chi^2 = 6984.3$, df=29, $p < 0.001$; Cramér's $V = 1.00$ with `hml12` (Fig. 1)
Imputation	Mean if $\lvert\text{skew}\rvert < 0.5$, else median; KNN if $> 5\%$ missing	Median for skewed, mean for Gaussian, constant "missing" for nominal; no column exceeds 5% except the structural pair; imputers keep indicators	13 right skewed, 5 other skewed, 12 Gaussian columns (Fig. 3)
Outliers	IQR rule for skewed, Z-score for normal; winsorize	Same, plus a percentile fallback (1st/99th) when IQR$= 0$	`Frost_Days_2020`: Q1=Q3=0, plain rule flags 2,986 rows (19.99%) and would flatten the feature to a constant (Fig. 3)
Association tests	Pearson/point biserial, Spearman or Mann–Whitney for skewed	Mutual information (MI) and point biserial with Spearman check; no switch needed	Largest $\lvert\rho\rvert - \lvert r\rvert$ gap is 0.013 (`hv040`: $\rho = -0.085$, $r = -0.072$); top MI only 0.016 nats (Fig. 6)
Collinearity	$\lvert r\rvert \geq 0.85$, then VIF> 10, for linear and distance models, optional for trees	Applied to the logistic branch only; trees keep all features	20 pairs with $\lvert r\rvert \geq 0.85$ (Figs. 4, 5); 12 features with VIF> 10; 10 dropped; max VIF falls to 7.54
Transformation	MinMax scaling	log1p for right skewed, then standardization	`Livestock_Goats` skew $4.12 \rightarrow -0.19$ (Fig. 7)
Encoding	Wealth quintile $\{0..4\}$, one hot ≤ 10 levels, target encoding for high cardinality	Explicit order, one hot with `drop=if_binary`, rare levels $< 1\%$ to "Other"; no high cardinality column used	22 nominal columns expand to the 95 model columns
Split	Group by cluster, stratify	StratifiedGroupKFold, seed chosen by label balance	Overlap 0; 143/36 positives
Imbalance	Class weights; SMOTE only if needed; threshold on validation	Class weights only; threshold on out-of-fold data (recall target 0.70)	103.5:1; SMOTE not needed
Calibration and weights	Planned	Platt and isotonic fit on out-of-fold predictions; weighted (`hv005`) and unweighted metrics	Brier $0.13 \rightarrow 0.0094$ for logistic (Table 3)
Model family and features	XGBoost, LightGBM or CatBoost on household plus climate features; logistic as baseline	Regularized logistic on household features plus cluster mean context is best out of fold; climate features lower PR-AUC (0.021 vs 0.041)	Pooled out-of-fold PR-AUC over 179 positives (Table 5)
Evaluation protocol	One 80/20 grouped hold-out	Hold-out kept, but headline numbers are pooled out-of-fold over all 18,672 rows	Hold-out has 36 positives from few clusters; one cluster holds 21.8% of positives
Climate source	CHIRPS, ERA5, MODIS buffer means	Main models: DHS 5 km covariates (2020 values). Added July to December 2023 interview month features: CHIRPS v3 (5 km buffer mean), ERA5-Land dewpoint, MODIS MOD13Q1 EVI and NDVI, lags 0 to 3	Pooled PR-AUC with them 0.011 to 0.021 vs 0.050 without (Table 6); ERA5 file held dewpoint only; deviation in Sec. 7

Table 2: Decision log: Phase 1 plan versus Phase 2 execution with the numerical justification.

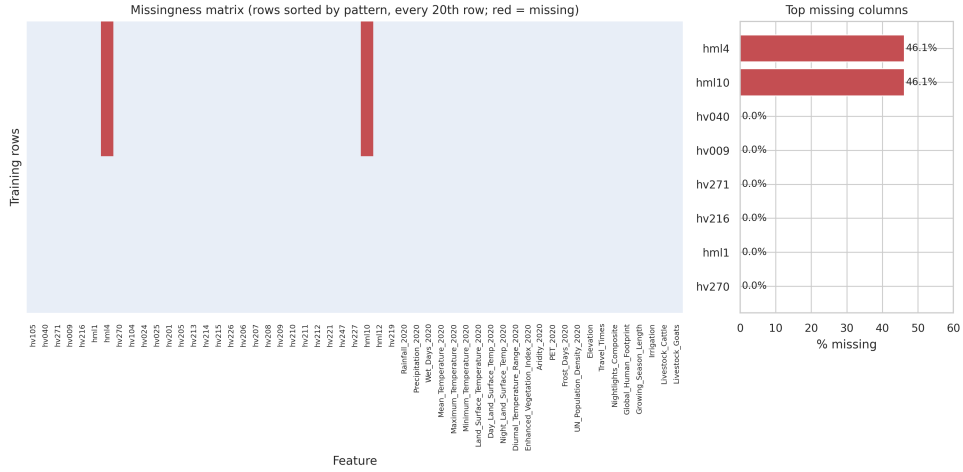


Figure 1: Missingness pattern (training rows sorted by pattern) and top missing columns.



Figure 2: Class balance by split (log scale) and age density by test result.

Model	PR-AUC	ROC	Brier	Recall
Logistic	.0132	.546	.0096	.53
Random forest	.0138	.625	.0096	.47
LightGBM	.0156	.656	.0097	.75
XGBoost	.0203	.643	.0096	.64
Baseline	.0096	.500		

Table 3: Hold-out test (3,732 rows, 36 positives), Platt calibrated, unweighted. Weighted PR-AUC: .0130, .0129, .0150, .0193.

positives its 95% interval $[0.005, 0.064]$ overlaps every other model.

Diagnosis. The weak scores are not a coding error. Out-of-fold and hold-out ROC-AUC differ (0.75 vs 0.64) because grouped splits remove cluster memorization, and the cause is the data. Only 58 of 170 clusters contain any positive; one cluster holds 39 of 179 positives (21.8%) and the top three hold 35.8%. Prevalence by province code ranges from 0.35% to 2.08%, by age from 0.68% to 1.22%. A 36 positive hold-out therefore reflects a handful of clusters. Climate covariates are constant within a cluster, so a flexible model can memorize the 170 distinct climate profiles, which does not transfer to unseen clusters. A model predicting which clusters contain a positive reaches ROC-AUC only 0.52 ± 0.03 (logistic) and 0.58 ± 0.03 (random forest).

Improved evaluation and model. To use all 179 positives, every person is predicted once by a model that never saw their cluster (5-fold cluster grouped, label stratified, pooled over 18,672 rows; Table 5). Five configurations were fixed in advance. Dropping the climate covariates and keeping 30 household features raises logistic PR-AUC from 0.021 to 0.041; adding cluster mean age, wealth score, household size, rooms and nets (label free context) raises it to **0.050** (ROC-AUC 0.72, weighted PR-AUC 0.048; the top 20% highest scores capture 49% of positives, a lift of 2.5). Tree models do not benefit, and adding all of MIS 2017 (1,765 positives, no climate) lowers logistic PR-AUC to 0.016, so survey round shift outweighs the extra positives. Training on 2017 alone and testing on all of 2023 gives PR-AUC 0.015 to 0.021. All intervals are wide (best model $[0.009, 0.137]$) and the five configurations were compared on the same data, so some optimism remains; the ranking of logistic over boosted trees is suggestive, not proven.

Table 4 summarizes the path: each retained step is justified by out-of-fold evidence, and the final step shows that a plausible improvement was tested and discarded rather than assumed.

Interview month climate features. After the first submission draft we downloaded CHIRPS v3 monthly rainfall (mean of pixels within 5 km of each displaced cluster point), ERA5-Land monthly 2 m dewpoint temperature (0.1° cell; the downloaded file contained dewpoint only, not tem-

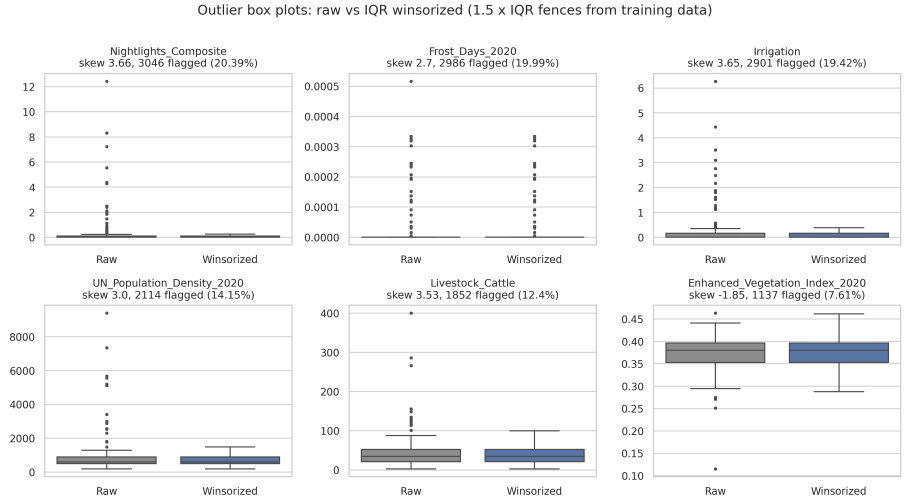


Figure 3: Outlier box plots, raw versus winsorized, for the six features with the most flagged values.

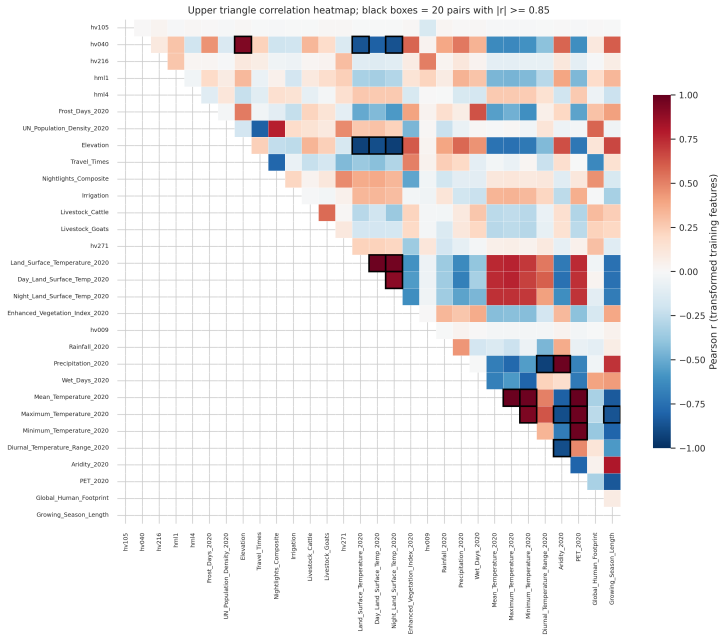


Figure 4: Upper triangle correlation heatmap of transformed numeric training features; boxes mark $|r| \geq 0.85$.

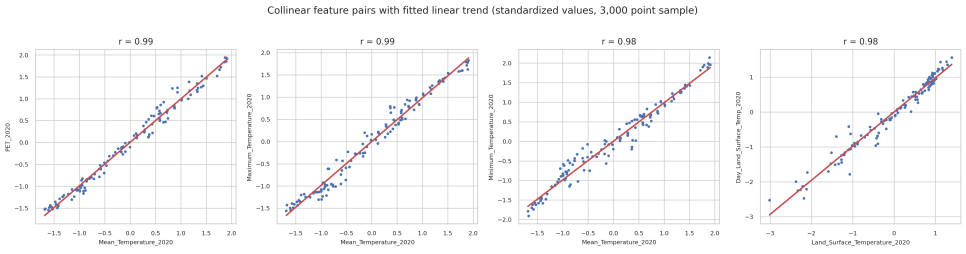


Figure 5: Collinear pairs with fitted linear trend (standardized values).

perature or rainfall) and MODIS MOD13Q1 EVI and NDVI composites (AppEEARS, quality filtered, 10.6% of interview month values missing after filtering). MIS 2023 interviews ran October to December 2023, so lags 0 to 3 months cover July to December 2023. Table 6 shows that adding them lowers every score: logistic PR-AUC falls from 0.041 (household) to 0.021, and boosted trees fall to near chance (0.011 to 0.013). Climate features alone give 0.012 (logistic) and 0.009

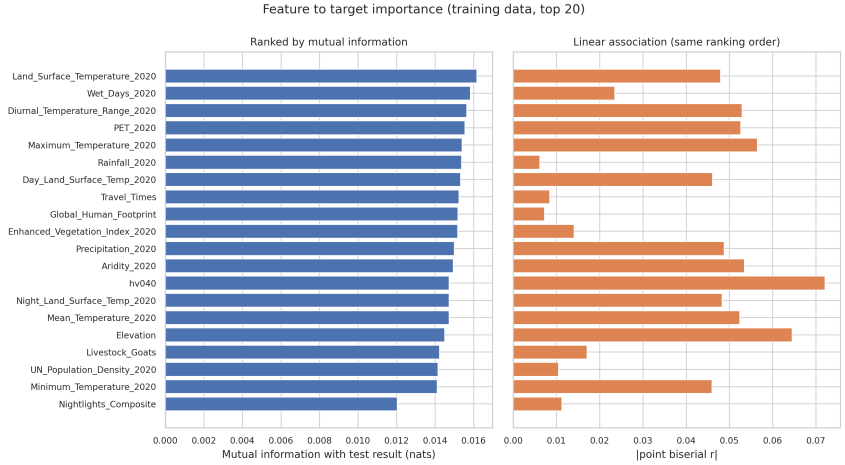


Figure 6: Feature to target importance: mutual information and |point biserial r| (training data, top 20).

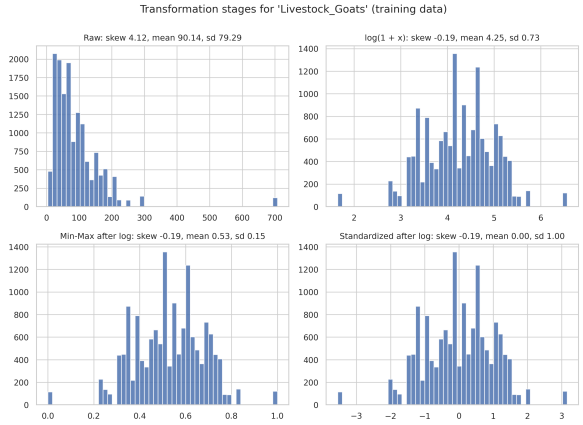


Figure 7: `Livestock_Goats`: raw, log1p, min-max and standardized.

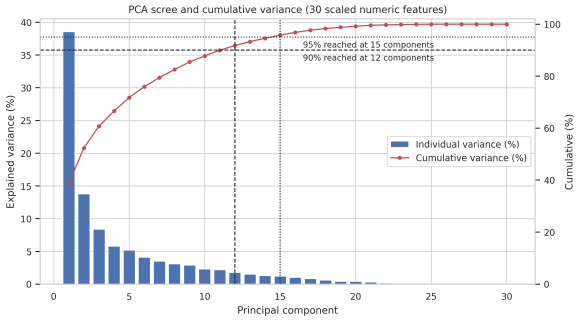


Figure 8: PCA scree and cumulative variance with 90% and 95% cutoffs.

(XGBoost, below the 0.0096 baseline). Only three interview months exist, so these features vary mainly between clusters; with 58 positive clusters, flexible models fit cluster specific noise that does not carry to unseen clusters. The result is a negative finding for climate at this resolution and sample size, not evidence that climate is irrelevant to transmission.

Step	Change	PR-AUC	Gain
Random baseline	none	.0096	1.0×
Hold-out, best (XGBoost)	36 positives	.0203	2.1×
Pooled, all features	179 positives, grouped CV	.0211	2.2×
+ household features only	drop climate	.0408	4.3×
+ cluster mean context	label free	**.0498**	**5.2**×
+ interview month climate	CHIRPS, ERA5, MODIS	.0194	2.0×

Table 4: Improvement path (logistic regression unless stated). The last row was tested and rejected.

Config	Model	PR-AUC	ROC	Lift@20%
A all feats	Logistic	.0211	.690	2.18
A all feats	XGBoost	.0162	.654	1.79
B household	Logistic	.0408	.717	2.57
B household	XGBoost	.0228	.693	2.26
C B + cluster means	Logistic	**.0498**	.717	2.46
C B + cluster means	LightGBM	.0189	.673	2.09
D B + MIS 2017	Logistic	.0159	.639	1.90
D B + MIS 2017	LightGBM	.0192	.676	1.87

Table 5: Pooled out-of-fold results over all 179 MIS 2023 positives (baseline PR-AUC 0.0096). Selected rows; all 15 model and configuration pairs are in `reports/pooled_evaluation.json`.

7 Limitations and Supervisor Feedback

We address four review points. (1) GPS displacement (urban up to 2 km, rural up to 5 km) is mitigated, not removed: the main models use DHS 5 km cluster covariates (2020 values). Interview month features from CHIRPS (5 km buffer), ERA5-Land and MODIS were added afterwards, but ERA5 supplied dewpoint only (0.1° grid, no buffer), MODIS values are 250 m pixel values

Features	Model	PR-AUC	ROC	Lift@20%
Household + dynamic	Logistic	.0213	.698	2.21
Household + dynamic	XGBoost	.0125	.585	1.37
HH + cluster means + dyn.	Logistic	.0194	.696	2.09
All static + dynamic	Logistic	.0136	.624	1.51
Dynamic only	Logistic	.0122	.593	1.33
Dynamic only	XGBoost	.0090	.450	0.92

Table 6: Pooled out-of-fold results with interview month climate (CHIRPS, ERA5-Land dewpoint, MODIS), 179 positives, baseline PR-AUC 0.0096. Best model without them: .0498 (Table 5).

with 10.6% missing after quality filtering, and only three interview months exist. (2) Normalized `hv005` weights are used in evaluation and both weighted and unweighted results are reported; weights are not used in training. (3) Calibration uses only out-of-fold predictions. (4) VIF reduction is restricted to the linear branch. Further limits: 179 positives in total, one survey round per model, and results that describe this survey only.

8 External Data Acquired and Still Needed

Table 7 records the external sources, what was obtained, and what remains. Acquired files live in `data/external/` and are git-ignored because provider and DHS terms prohibit redistribution.

9 Conclusion

The pipeline is leakage free and fully auditable, but malaria positivity in MIS 2023 is dominated by a few clusters that household and coarse climate covariates only partly locate. The honest finding is a modest but real lift: the best out-of-fold model (regularized logistic regression on household features plus cluster context) ranks positives about five times better than chance in PR-AUC terms, with wide intervals. Interview month climate from CHIRPS, ERA5-Land and MODIS did not add skill at this sample size; more survey rounds with GPS, ERA5 temperature and rainfall, and district case series (Table 7) are the next steps.

Data	Source	Purpose and status	Status
Monthly rainfall, lags 0 to 3, 5 km buffer around each GPS cluster	CHIRPS v3 [9], Climate Hazards Center	July to December 2023 downloaded and joined by cluster and interview month (`src/climate_features.py`)	Acquired (2023)
Dewpoint temperature, monthly	ERA5-Land [10], Copernicus CDS	2017 and 2023 files hold `d2m` only; 2 m temperature and total precipitation were not in the files and must be requested again	Partial
Vegetation (EVI, NDVI), 16 day	MODIS MOD13Q1 v061 [11], via AppEEARS	170 clusters, 12 composites each, with quality flags; joined as monthly lags	Acquired
Population denominators and district boundaries	NISR census 2022 [13]	District roll-up of cluster risk and exposure normalization	Pending
Monthly district malaria cases, 2014 to 2024	Rwanda Biomedical Centre bulletin [12]; full series by formal request	Enables the time series early warning in the original proposal and independent validation	Pending
Rwanda DHS 2019–20 and MIS 2017 GPS	DHS Program (separate requests)	Third survey round (1.79% RDT prevalence) and climate features for 2017	Pending

Table 7: External data: acquired versus still needed.

A Original Project Proposal

This appendix reproduces the original proposal text unchanged, including its original scope (district level monthly early warning from surveillance data). Section 7 and Table 2 explain how and why the scope changed to individual level RDT classification.

Climate Informed Machine Learning for District Level Malaria Early Warning in Rwanda

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Abstract

Malaria remains a leading cause of morbidity in Rwanda despite gains achieved through vector control and treatment programs. District level surveillance recorded through the Rwanda Biomedical Centre Health Management Information System is used mainly to describe past trends rather than anticipate surges, and annual net and spraying allocations ignore short term climate signals. This proposal presents a climate informed machine learning system that forecasts district level malaria risk one to two months ahead across three ordinal levels, normal, elevated, and epidemic, so that officials can position bed nets, spray teams, and drug stocks before, rather than after, a seasonal peak. The system frames prediction as multiclass classification, in which gradient boosted trees and a multinomial logistic regression baseline learn from lagged rainfall, temperature, humidity, vegetation wetness, and case history. Three pilot districts are selected by historical case burden and rainfall driven variability, with additional districts held out to test regional transferability. Validation follows a temporal split that holds out complete malaria seasons and is assessed with balanced accuracy, macro F1 score, and a lead time detection rate, benchmarked against a seasonal persistence baseline. This work directly supports Sustainable Development Goal 3, specifically the target of ending the epidemics of malaria, by translating openly available climate, vegetation, and surveillance data into an operational early warning tool for a resource constrained health system.

A.1 Introduction and SDG Relevance

Malaria continues to threaten public health in Rwanda even though control programs have driven major reductions in transmission over the past decade [14]. A public health bulletin covering 2014 through 2024 shows that incidence and severity among children under five and pregnant women have fluctuated rather than declined steadily [12]. Surveillance through the District Health Management Information System is compiled after cases have already occurred, so district teams receive information that describes the past rather than a signal that anticipates the near future. Rainfall, temperature, humidity, and vegetation wetness are documented drivers of Anopheles mosquito breeding and Plasmodium falciparum development, and recent studies across Africa show that climate covariates carry predictive information for malaria months ahead of an outbreak [15, 16, 17]. This project combines openly available satellite rainfall, temperature, humidity, and vegetation records with historical district level surveillance data to build a machine learning early warning system for Rwanda, turning existing open data into a forecasting tool that district teams can use to plan interventions before, rather than after, a seasonal rise in cases.

A.2 Selected SDG

SDG 3, Good Health and Well Being, target 3.3, calls for ending the epidemics of malaria by 2030. The proposed system forecasts district level malaria risk so the national malaria program can allocate long lasting insecticidal nets, indoor residual spraying teams, and antimalarial drug stocks to the districts and months where an elevated or epidemic case load is expected, supporting progress toward elimination rather than only measuring it after the fact.

A.3 Refined Problem Statement

The Ministry of Health currently monitors malaria through retrospective descriptive statistics drawn from the District Health Management Information System, aggregated by district and year [12]. This accurately documents what happened but does not forecast what is likely to happen. Districts with a historically disproportionate case burden are also subject to strong rainfall driven transmission cycles, yet planning cycles for net distribution and spraying are set annually rather than adjusted to

short term climate signals, so districts can be under supplied ahead of a severe season or over resourced ahead of a mild one. The deficiency is not a lack of surveillance data but an absence of a predictive layer that converts existing rainfall, temperature, humidity, vegetation, and case history into a forecast with enough lead time, typically one to two months, for procurement and distribution logistics to respond. The initial pilot targets the three districts with the highest documented case burden and the strongest rainfall driven case variability reported in the national malaria bulletin [12], and additional districts spanning other agroecological zones will be scored with the same trained models to test regional transferability before any broader rollout is considered. Without such a layer, preventable cases concentrated among children under five and pregnant women continue to occur even where nets, drugs, and spraying capacity already exist within the health system.

A.4 AI Task Formulation

The prediction task is a monthly, district level ordinal classification problem. For district d and target month t, let $Y(d,t)$ take the value zero for normal incidence, one for elevated incidence, and two for epidemic incidence, defined respectively as below the historical seventy fifth percentile, between the seventy fifth and ninetieth percentile, and above the ninetieth percentile for that calendar month. This extends the above versus below median formulation used in prior work [15] with an epidemic tier that logistics planners can act on directly. The feature vector $X(d,t)$ combines the most recent one to three months of rainfall, temperature, humidity, and vegetation wetness observations, the district's own case counts from the previous one to three months, and calendar month as a categorical feature to capture residual seasonality. The task is supervised multiclass classification, $f : X \to Y$, trained separately for lead times of one and two months. A multinomial logistic regression baseline with L2 regularization is compared against a gradient boosted tree ensemble, since gradient boosted trees achieved the highest reported accuracy among classifiers compared in a recent malaria study [17] and natively support multiclass outputs. Hyperparameter tuning uses nested cross validation with a chronological split so no future month predicts an earlier one, and the two most recent complete malaria seasons are held out as a final test set.

A.5 Initial Data Sources

- Rwanda Biomedical Centre DHMIS, via the RBC Public Health Bulletin: monthly and yearly malaria case counts, deaths, and severity by district, 2014 through 2024 [12]. https://rbc.gov.rw/publichealthbulletin/
- WHO Global Health Observatory and World Malaria Report Rwanda country profile, for cross checking national trends [18]. https://www.who.int/publications/m/item/malaria-2024-rwa-country-profile
- CHIRPS rainfall data, Climate Hazards Center, UC Santa Barbara: daily gridded rainfall at 0.05 degree resolution, 1981 to near present [9]. https://www.chc.ucsb.edu/data/chirps
- CHIRTS daily temperature dataset from the same center, accessed with the same retrieval tools. https://www.chc.ucsb.edu/data
- ERA5 Land hourly reanalysis, Copernicus Climate Data Store: relative humidity and dewpoint temperature, which influence adult mosquito survival beyond rainfall alone [10]. https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land
- MODIS sixteen day vegetation index product, NASA Land Processes Distributed Active Archive Center: a proxy for surface water accumulation and breeding habitat [11]. https://lpdaac.usgs.gov/products/mod13q1v061/
- NISR Fifth Population and Housing Census, for district population denominators [13]. https://statistics.gov.rw/

Full monthly case series for all thirty districts, beyond the aggregated bulletin figures used here, will require a formal request to the Rwanda Biomedical Centre.

A.6 Expected Challenges

- Short historical record and possible reporting changes in the DHMIS over time: eleven years of monthly data limits sample size, so districts will be pooled by agroecological zone before specializing per district, and national totals will be cross checked against WHO figures [18].

- Coarse climate grid relative to district boundaries and class imbalance across the three risk levels: address with area weighted averaging of pixels per boundary and with balanced class weights, reporting balanced accuracy and macro F1 rather than raw accuracy.
- Cross border movement, insecticide resistance, and past intervention timing, including spraying and net distribution: none are yet available as structured records, so they are treated as residual error in this pilot, with intervention history to be requested from the Malaria and Other Parasitic Diseases Division for later use as a covariate.
- Limited district connectivity: run a lightweight, centralized monthly scoring script rather than a real time streaming system.

A.7 Preliminary Success Metrics

Performance is measured with balanced accuracy, macro F1 score, and area under the ROC curve at each lead time, compared against a seasonal persistence baseline that repeats the historical class for each calendar month. An operational lead time detection rate, defined as the proportion of elevated and epidemic months correctly flagged at least one month before onset, will be reported alongside these conventional metrics since it more directly reflects the reduction in stockout risk that the system is intended to support. A preliminary target of seventy five percent balanced accuracy at a one month lead is proposed, to be refined once exploratory analysis of the actual class balance and noise level is complete. Predicted probabilities will also be checked for calibration with a reliability plot, and a short structured feedback session with malaria program staff will assess the usability of the forecast bulletin format.

A.8 Project Timeline and Deployment Resources

- Weeks one to four: problem refinement, literature review, data acquisition across all sources, district boundary mapping, and cleaning of surveillance and climate records.
- Weeks five to seven: exploratory analysis, lag correlation between covariates and case counts, and development of the multinomial logistic regression baseline.
- Weeks eight to ten: gradient boosted tree model, nested cross validation, hyperparameter tuning, and held out evaluation against the persistence baseline, with the final days of week ten reserved as a contingency buffer to revisit model choice if performance does not clearly exceed the baseline.
- Weeks eleven and twelve: prototype monthly bulletin, packaging of the scoring script, documentation, and final presentation.

Deployment resources are entirely open source: Python with scikit learn and XGBoost for modeling, retrieval clients for the climate and vegetation archives listed above, geopandas for district boundary handling, a lightweight SQLite database for monthly features, Docker for packaging the scoring script, and a scheduled monthly job to regenerate the bulletin. The extracted dataset is small enough to train on a standard laptop, and a future pilot could run on existing Ministry of Health infrastructure or a modest cloud instance to keep retraining costs low.

Limitations

This pilot uses a single national surveillance stream and openly available gridded climate and vegetation data, so it cannot yet capture cross border population movement, vector insecticide resistance trends, or the timing of past indoor spraying and net distribution campaigns, all of which independently affect malaria incidence and are not currently available to this project as structured historical records. The eleven year monthly record available per district is short by contemporary machine learning standards, which restricts model complexity and requires cautious interpretation of accuracy figures until more years of data accumulate. Predictions are generated at district resolution and are not intended to guide clinical decisions for individual patients. The coarse resolution of the climate grid relative to district boundaries also introduces averaging error that finer resolution satellite products could reduce in future work.

Ethics Statement

All data sources proposed for this project are aggregate and publicly released, so no individual patient records are accessed and no additional informed consent process is required beyond what each data provider already applies. Because the output is a probabilistic forecast intended to support, not replace, expert judgment within the national malaria program, the system will present uncertainty alongside each prediction and will be described to end users as decision support rather

than an automated allocation mechanism, reducing the risk that a false negative forecast leads to under preparation in an at risk district. Any future extension toward facility level data or historical intervention records would require explicit data sharing agreements with the Rwanda Biomedical Centre and adherence to applicable data protection law.

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